POL583: Estimating the effect of Super PACs on minority representation in local elections

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Abstract

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Contents

Introduction

In the wake of the 2010 Citizens United v. Federal Election Commission ruling, Super Political Action Committees (Super PACs) have become a defining feature of American politics, enabling corporations and unions to make unlimited independent expenditures. While proponents argue that this ruling amplifies political participation, critics worry it disproportionately empowers wealthy donors, thereby diminishing equitable representation. These concerns are particularly acute in local elections, where minority candidates often serve as a vital link between underrepresented communities and the political process.

The influence of wealthy donors on local elections is not merely theoretical—it is vividly illustrated by cases such as Amazon's involvement in Seattle's 2019 City Council elections. Facing regulatory threats, Amazon contributed an unprecedented 1.5 million dollars to the city's chamber of commerce PAC, which backed pro-business candidates. Although the majority of Amazon-supported candidates ultimately lost, the controversy underscored a larger concern: the growing ability of corporations and affluent individuals to shape local electoral outcomes in their favor. Anecdotal evidence like this suggests that the rise of Super PACs amplifies the voices of the wealthy while sidelining underrepresented groups, particularly women and non-White candidates, who already face systemic barriers to political participation.

This paper moves beyond anecdotal cases to provide a systematic analysis of how Super PACs affect minority representation in local elections. It tests three hypotheses: first, that Super PACs reduce the probabilities of minority candidates existing, entering, and winning local contests (H1); second, that these effects are more pronounced in non-partisan elections, where campaign finance is typically smaller and less transparent (H2); and third, that non-White candidates experience greater reductions in representation than female candidates due to the intersecting effects of racial and economic inequality (H3).

To address these questions, this study employs a dataset encompassing over 43,000 local elections across 57,000 contests spanning a decade before and after the Citizens United ruling. Using a difference-in-difference (DiD) two-way fixed effects model, the analysis disentangles the causal effects of Super PACs on minority representation, with a particular focus on subgroup heterogeneity and partisan versus non-partisan contexts. By systematically exploring the dynamics of minority representation in the era of Super PACs, this paper addresses concerns raised by high-profile cases like Amazon's influence in Seattle, offering broader insights into how corporate money reshapes the democratic landscape of local elections—and whose voices are amplified or silenced.

Literature Review

Rise of super PACs

Super PACs, or "independent expenditure-only committees," emerged in the wake of two pivotal U.S. court decisions: the 2010 Citizens United v. Federal Election Commission and the SpeechNow.org v. FEC rulings. These decisions removed restrictions on corporate, union, and individual spending for political advocacy, provided such expenditures are made independently of candidates' campaigns. Unlike traditional PACs, super PACs can raise unlimited funds from wealthy individuals, corporations, and unions and spend them to influence elections, though they cannot directly coordinate with candidates. This development revolutionized campaign financing, enabling unprecedented levels of outside spending in elections and intensifying concerns about the influence of wealthy donors and business interests on the democratic process.

Wealthy donors and business interests

There exist many studies on the demographics and policy views of donors. Donors are older, higher-income, wealthier, more likely to be male and less likely to be a racial and ethnic minority Grumbach and Sahn (2020). Donors are more ideologically extreme that non-donors (Barber et al., n.d.).

Based on the two stylized facts that donors are more affluent and that donors are more ideologically extreme, scholars of campaign finance have assumed that the affluent are more ideologically extreme than the general public. However, a recent study by Barber et al. (n.d.) found that wealthy donors are as ideologically extreme as donors, but wealthy non-donors are not.

Unlike wealthy donors, business interests have been portrayed as less expressive than individual donors by the literature of campaign finance, based on the fact that many business interests contribute to multiple candidates in each election (Barber 2016). Their primary goal has been identified as seeking access to policymakers rather than supporting their ideological allies (Ansolabehere, Figueiredo, and Snyder 2003; Kalla and Broockman 2016).

Studies analyzing the policy preferences of wealthy donors or business interests are based on surveys or observational campaign contributions data from the Federal Election Commission (FEC) (Barber et al., n.d.; Bonica 2014). A common key limitation on both types of data is that they are mostly related to federal policy. Surveys ask respondents about their national policy preferences. FEC campaign contributions data is from those contributing at the federal level. Donors or PACs participating in state or local elections only do not appear in the database. Donors and PACs contributing to both federal and lower-level elections do appear in the database, but their contributions to lower-level elections do not appear in the database. Therefore, donors' and business interests' ideologies derived from FEC campaign contributions data only speaks to their broad national policy preferences.

Local vs upper-level elections

Wealthy donors and business interests' ideologies, derived from their broad national policy preferences, do not predict their local policy preferences, as national and local politics cover significantly different issues. While national politics cover many ideological issues such as abortion, gun control, and police reform, local politics focuses on practical and immediate concerns such as public safety, housing affordability, and road maintenance. Therefore, local elections are less partisan and ideologically-driven than upper-level elections. [@]

Measuring wealthy donors and business interests' local policy preferences is an extremely difficult task. The first challenge is creating the survey questionnaire itself. Since different localities have different policy agenda, it is nearly impossible to collect policy agenda from all localities and distill them into a reasonable length survey. Moreover, since wealthy donors account for an extremely small subset of population, they are hard to reach. A typically low survey response rate may not yield a reasonable amount of responses to call opinion of wealthy donors and business interests.

However, the literatures on local politics and business interests provide predictions on wealthy donors and business interests' local policy preferences. The literature on local politics says that local politics are mostly non-partisan and driven by economic concerns. Considering that the U.S. remains significantly segregated by both income and race, wealthy donors are very likely to be close neighbors in white neighborhoods and share local policy preferences, such as where to build low-income apartment housing (Fry and Taylor 2012; Loh and Buthe 2020). The literature on business interests say that although business interests might have different preferences on cultural dimension, they share the same goal of business enhancement (GILENS, PATTERSON, and HAINES 2021). Then, with these local policy preferences, how do wealthy donors decide which candidate to support in predominantly non-partisan local elections?

Race and gender in local elections

Race in local elections

The racial background of candidates plays an important role in local elections (Boudreau, Elmendorf, and MacKenzie 2019; Benjamin 2017; Beach et al. 2024). In diverse communities, a candidate's racial background may hold greater significance as voters may seek representation that aligns with their own identity or community concerns. Local elections often focus on addressing specific community needs, which can make racial identity a focal point. Considering the racialized nature of local elections, wealthy donors are expected to support candidates from the same white racial background.

Gender in local elections

The gender of candidates plays a more nuanced role at the local level. Research on gender dynamics in political donations indicates that male donors predominantly contribute to male candidates, while female donors tend to support both male and female candidates. Thomsen and Swers (2017) found that female candidates, particularly within the Republican Party, often face challenges in fundraising due to limited access to donor networks, which are predominantly male. This limitation suggests that male donors may prefer male candidates, whereas female donors are more inclusive in their support. Additionally, Sorensen and Chen (2022) found that women of color are systematically disadvantaged in campaign finance, receiving less funding compared to their white and male counterparts. This disparity underscores the influence of donor demographics on candidate support. These studies suggest that male donors may exhibit a preference for male candidates, while female donors distribute their contributions more equitably across genders. This dynamic contributes to the fundraising challenges faced by female candidates, particularly those from minority backgrounds.

However, local offices often deal with community-focused issues like education or public services, areas where female candidates are sometimes perceived as having strengths due to gender stereotypes. Anzia and Bernhard (2022) found that female candidates have greater advantages over men in school board and city council than mayoral races. The advantages are smaller in conservative constituencies. Thus, female candidates fare better in contexts where gender stereotype is congruent and worse in contexts where incongruent.

To summarize, wealthy donors are more likely to support candidates of the same race and gender in local elections. However, in some elections for offices where female candidates are perceived as having advantages such as school board and city council elections, female candidates have greater advantages than their male counterparts.

How super PACs affect local elections?

Assuming white and male wealthy donors prefer candidates of the same race and gender in local elections, what would they do to elect the candidates they want in the super PAC era? Little systematic analysis exist on super PACs' impact on local elections. The vast majority of studies quantitatively measuring the effect of super PACs were at the state level. Such study at the federal level is impossible due to a single case study problem. Measuring causal effects in presidential elections is challenging because there is only one observation of the outcome: the election of a single president. Since there is only one outcome, it is difficult to use traditional statistical methods to analyze the impact of various factors on the result.

Such study at the local level has been impossible due to a lack of comprehensive U.S. local elections database. However, Benedictis-Kessner et al. (2023) has made such a database - American local government elections database - available in 2023. This database includes 78,000 candidates in 57,000 electoral contests that encompasses races for seven distinct local political offices in most medium and large cities and counties in the U.S. over the last three decades. For more details about this database, please check the data section under Research Design.

Despite the availability of this database, another challenge still exists in studying the effect of super PACs on local elections: a lack of comprehensive super PACs data. Super PACs exist at different levels of elections. Super PACs at the federal level focus on influencing congressional races, Senate elections, and presidential campaigns. These PACs are often well-funded, ideologically driven, and target national policies. At the state level, super PACs can influence gubernatorial elections, state legislative races, and statewide ballot initiatives. These PACs often align with regional policy interests or focus on state-specific issues, such as healthcare, education, or infrastructure. Local super PACs influence city council elections, mayoral races, school board elections, judicial contests, and local ballot measures. They bring external resources to campaigns that may otherwise rely on smaller, grassroots fundraising efforts.

Unlike federal super PACs, state and local super PACs are not required to follow FEC filings, as the FEC primarily oversees campaign finance for federal elections, including campaigns for

President, Senate, and House of Representatives. Instead, state and local super PACs must adhere to the campaign finance laws and regulations of their respective states or jurisdictions. The degree of regulation varies widely by state and locality.

Due to a lack of uniformity in campaign finance laws across different states and localities, creating a comprehensive super PAC database is a non-trivial task. For this reason, most scholars of campaign finance use the FEC data, confining their analysis to the federal level. Creating a comprehensive federal, state, and local super PACs database is beyond the scope of this paper.

Due to a lack of comprehensive super PACs data including local super PACs, no systemic analysis exists on how wealthy donors influence local elections. However, anecdotal evidence abound. In 2019, Amazon contributed 1.5 million dollars to the super PAC ran by the Seattle Metropolitan Chamber of Commerce to support seven corporate-friendly city council candidates. That amount accounted for more than half of nearly 2.7 million dollars raised by the super PAC. In 2021 New York mayoral election, several single-candidate super PACs raised about 7.3 million dollars in total (Rubinstein and Mays 2021). Candidate Shaun Donovan's father, Michael Donovan, established a single-candidate super PAC for his son. The super PAC raised more than 3 million dollars, nearly all of it from his father. A recent investigation by Guardian shows that tech and venture capital leaders in San Francisco, including the PayPal co-founder David Sacks and the Pantheon CEO Zachary Rosen, have invested at least 5.7 million dollars into reshaping San Francisco's policies by creating a network of interlocking non-profits, dark money groups, and political action committees (Winston, Jarrett, and Local 2024). The most prominent group is the Neighbors for a Better San Francisco Advocacy, a local super PAC primarily funded by a handful of extremely wealthy donors from the tech and real estate.

Combining the theoretical expectation that wealthy donors prefer candidates of the same race and gender and the anecdotal evidence that wealthy donors pour money to local super PACs to influence local elections, the rise of super PACs are expected to reduce minority representation in local elections.

Hypotheses

Assuming that wealthy donors prefer candidates of same race and gender, I construct my first hypothesis as below:

H1: Super PACs decrease the probabilities of minority existing, entering, and winning in local elections.

How would the effect size differ by whether election is partisan or not? Partisan elections are generally higher profile with larger campaign finance. In a high-profile election with larger campaign finance, the influence of a handful wealthy donors is smaller than a low-profile election with smaller campaign finance. Based on this reasoning, I construct my second hypothesis as below:

H2: The effect size is larger for non-partisan local elections.

How would the effect size vary by different subgroup of minority? Combining two stylized facts that the U.S. remains significantly segregated by both income and race and that female candidates have more advantages over male candidates in some sterotype-congruent local elections, I theorize that wealthy donors, who are whiter and more male, prefer white female candidates over non-White candidates. Based on this reasoning, I construct my third hypothesis as below:

H3: The effect size is larger for non-White candidates than female candidates.

Research design

Data

The main data source is American local elections database released in late 2023 by Benedictis-Kessner et al. (2023). This database include about 78,000 candidates in 57,000 electoral contests that encompasses races for seven distinct local political offices in most medium and large cities and counties in the U.S. over the last three decades. The seven local political offices are city council, county executive, county legislature, mayor, prosecutor, school board, and sheriff.

I filtered the data so that it spans 10 years before and after the 2010 ruling. Elections are not held every year, so it is important to have a generous time span to gather multiple data points for each distinct contest. This step left 43,498 local elections. Among these local elections, 53 percent had at least one minority candidate (40 percent had at least one female candidate, and 30 percent had at least one non-White candidates). 41 percent were partisan elections, meaning there existed at least one Republican and one Democratic candidates. The breakdown by office is: 39 percent county legislatures, 32 percent city council, 9 percent prosecutor, 7 percent sheriff, 6 percent mayor, 5% school board, and 1 percent county executive.

Methodology

Model

I use the difference-in-difference (DiD) two-way fixed effects (TWFE) model. The model includes contest fixed effects, which account for time-invariant characteristics that might confound the treatment effect. It also includes time fixed effects, which control for trends applied to all units over time, such as national political changes.

$$Y_{ct} = \beta CityUni_t * BanState + Contest_c + Year_t + e_{ct}$$

Operationalization

The treatment variable is the product of *CityUni* and *BanState*. *CityUni* is a binary variable that indicates whether the year is after the 2010 Citizens United v. Federal Election Commission. The value is 1 if the year is above 2010 and 0 if below. *BanState* is a binary variable that indicates whether the state had a ban on independent expenditures by unions and/or corporations before 2010. Eight states that had a ban on expenditures by corporations only are Connecticut, Iowa, Kentucky, Massachusetts, Minnesota, Montana, Tennessee, and West Virginia. Fifteen states that had a band on expenditures by both corporations and unions are Alaska, Arizona, Colorado, Michigan, New

Hampshire, North Carolina, North Dakota, Ohio, Oklahoma, Pennsylvania, Rhode Island, South Dakota, Texas, Wisconsin, and Wyoming. In sum, the 23 states mentioned here have a value of 1 and the remaining states have a value of 0 for *BanState*. The product of *CityUni* and *BanState* is 1 only when both values are 1. The construction of this treatment variable is inspired by Abdul-Razzak, Prato, and Wolton (2020).

I use four binary outcome variables: *MinExists*, *MinEnters*, *MinWins*, and *MinWinsWhenRunning*. *MinExists* indicates whether a minority candidate exists in the race, regardless of whether the candidate is incumbent or new. *MinEnters* indicates whether a new minority candidate enters the election. *MinWins* indicates whether a minority candidate wins the race. However, this is not conditional on a minority candidate running. On the other hand, *MinWinsWhenRunning* indicates whether a minority candidate wins the race conditional on any minority candidate existing in the race. Here, minority candidates include both female and non-White candidates.

The unit of analysis is contest. Contest is a combination of state, locality, district, and office. An example is City of Los Angles Council District 10. It is a combination of the State of California, the City of Los Angles, Office of City Council, and District 10. For city-wide elections such as mayoral elections, contest is a simpler combination of state, locality, and office. The coefficient of interest β estimates the effect of the lift of corporate or union expenditure ban on the four outcome variables for each contest. The coefficient is derived from the variation of the outcome variables within the same contest.

A unique challenge for any time-series analysis of local political districts is redistricting. Larger cities with multiple districts are subject to redistricting, which happens typically every 10 years, following the release of the U.S. Census data. After redistricting the boundary for the same district - City of Los Angles Council District 10, for example - can be greatly different from before.

Excluding elections subject to redistricting eliminate important legislative elections of large cities from the sample, which are arguably more important due to their large population size. Ignoring redistricting and assuming districts remain the same makes it impossible to disentangle the effect of redistricting and the effect of super PACs.

A possible solution that includes many important districts without ignoring redistricting is to include only districts that saw minimal change in their boundaries. Many districts remain the same after redistricting, although it is hard to know whether they remain the same due to incumbents' political manipulation or insignificant population change. I define districts that saw minimal boundary change as those whose area change less than 5 percent. Identifying those districts is possible by overlapping maps before and after redistricting in GIS software such as ArcGIS.

Please note that due to the time constraints, my preliminary results does not address the issue of redistricting. My preliminary analysis assumes redistricting does not exist and boundaries of all districts remain the same throughout.

Preliminary Results

Figure 1 shows the effects of super PACs on minority representation in both partisan and non-partisan local contests. *Min* indicates the average effects on minority candidates, which include both female and non-White candidates.

Values for *Min* show that super PACs had statistically significant effects on all four outcome variables for minority candidates. The rise of super PACs had negative, statistically significant effects on the probabilities of minority candidates existing, entering and winning in a race. The effect on the probability of minority candidates entering was larger than that on the probability of minority candidates existing. Such a result makes sense considering that minority incumbents who already established their positions are likely to be impacted less than new minority candidates.

Super PACs had a negative effect on the probability of minority winning and a positive statistically significant effect on the probability of minority winning conditioning on minority running. I interpret the fact that super PAC's positive effects on minority conditioning on minority running as that super PACs create such a great barrier for minority candidates that only extremely qualified minority candidates can run. At this moment, I do not have local candidate quality data to prove this point.

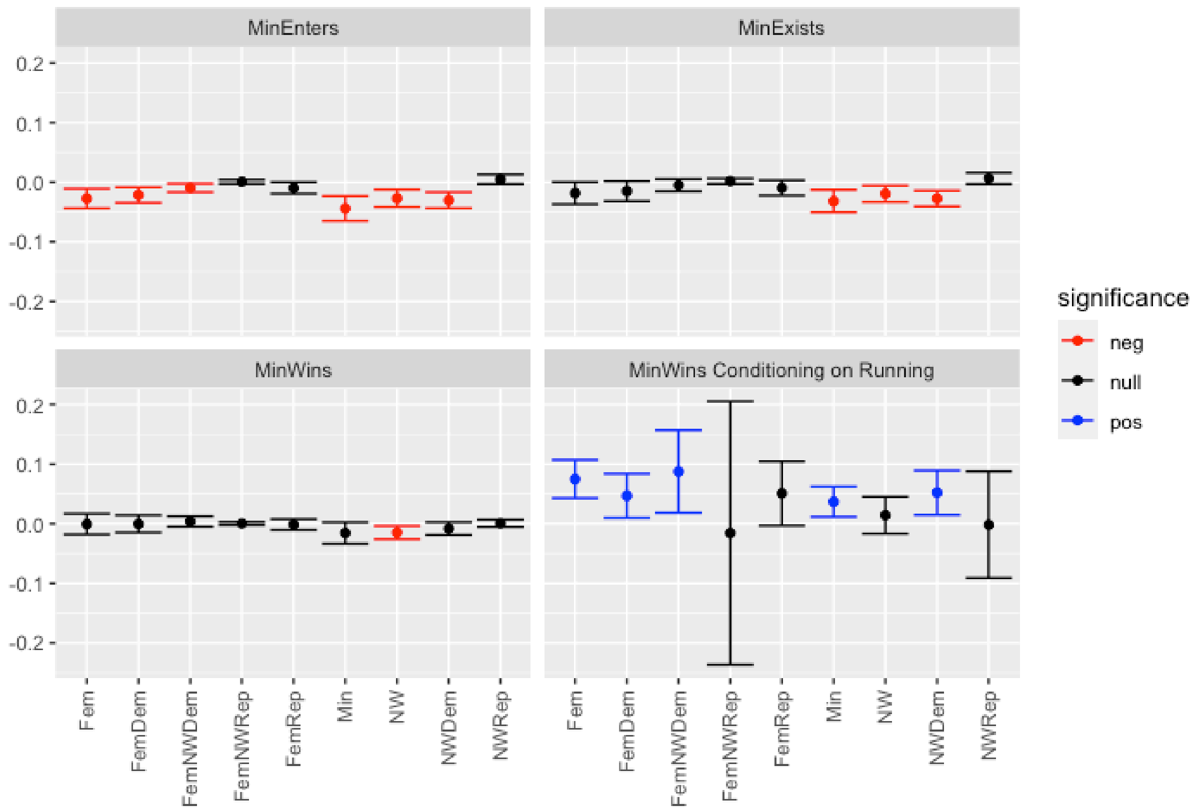

Figure 1: Effect of super PACs on minority representation in both partisan and non-partisan local contests. The Y-axis indicates the log-odds ratio from the model. The X-axis indicates the type of minority candidates. From left to right, *Fem* indicates female candidates. *Dem* indicates Democratic candidates. *NW* indicates non-White candidates. *Min* indicates minority candidates, which include both female and non-White candidates. *Rep* indicates Republican candidates. The black dots represent point estimates, and the error bar shows the 95 percent confidence interval. The blue error bars indicate positive and statistically significant effects. The red error bars indicate positive and statistically significant effects.

The heterogeneous treatment effect analysis shows that the effects are driven by specific types of minority candidates. Super PACs did not have statistically significant effects on female Republican candidates' probabilities of existing, entering, and winning. However, they had statistically significant effects on female Democratic candidates' probabilities of entering, and winning. Super PACs reduced Democratic candidates' probability of entering and increased their probability of winning.

The worst victims seem to be non-White candidates. Super PACs had a negative and statistically significant effects on non-White candidates' probabilities of existing, entering, and winning not conditional on minority candidate running and insignificant effect on their probability of winning conditional on minority candidate running.

This is unique because all subgroups, except for non-White candidates, that saw negative and statistically significant effects on their probability of entering also saw positive and statistically significant effects on their probability of winning conditional on minority candidate running. I interpret such a pattern as super PACs increase the barrier for minority candidates and allow only extremely qualified minority candidates to enter. However, non-White candidates are the exception of the pattern. One possible interpretation could be that even the most qualified non-white candidates who overcame the entrance barrier fall short of winning. My analysis does not explain why non-White candidates are an exception.

Figure 2 shows the effects of super PACs on minority representation in partisan local contests only. Values for *Min* show that super PACs had statistically significant effects on two of the four outcome variables for minority candidates. The rise of super PACs had negative, statistically significant effects on the probabilities of minority candidates existing, and entering and insignificant effects on the probabilities of minority winning, regardless of conditioning on minority candidate running.

The only difference from the analysis above that also includes non-partisan contests is on the probability of minority winning conditional on minority running. In the analysis that also includes non-partisan contests, super PACs had a positive statistically significant effect on the probability

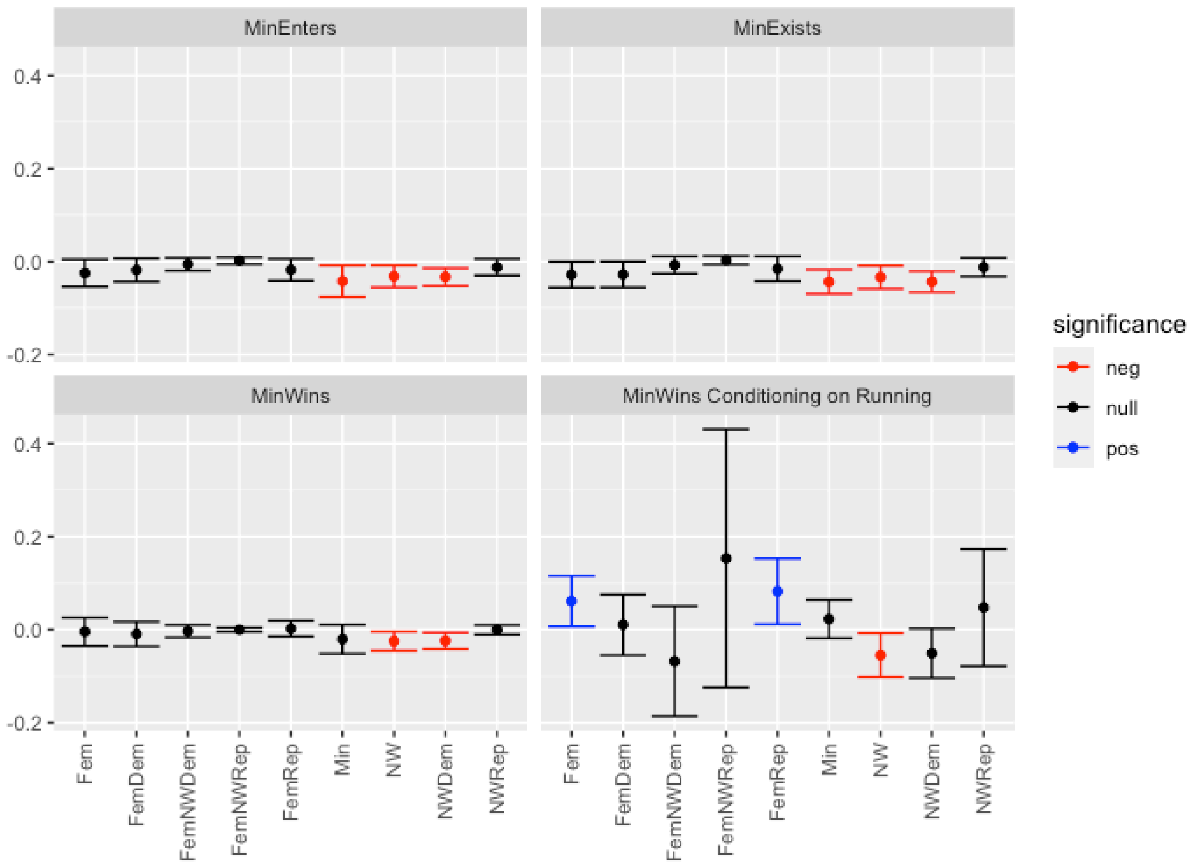

Figure 2: Effect of super PACs from partisan local contests. The Y-axis indicates the log-odds ratio from the model. The X-axis indicates the type of minority candidates. From left to right, *Fem* indicates female candidates. *Dem* indicates Democratic candidates. *NW* indicates non-White candidates. *Min* indicates minority candidates, which include both female and non-White candidates. *Rep* indiciates Republican candidates. The black dots represent point estimates, and the error bar shows the 95 percent confidence interval. The blue error bars indicate positive and statistically significant effects. The red error bars indicate positive and statistically significant effects.

of minority winning conditional on minority running. However, in the analysis that only includes partisan contests, super PACs had an insignificant effect on the probability of minority winning conditional on minority running.

Why is that so? The heterogeneous treatment effect analysis provides an answer. The heterogeneous treatment effect analysis clearly shows who are the beneficiaries and victims of super PACs. Female Republicans benefited from super PACs compared to other minority candidates. Super PACs did not have any statistically significant effects on Female Republicans' probabilities of existing, and entering, but had a positive statistically significant effect on their probability of winning conditional on female Republican running. Unlike female Republicans, female Democrats did not enjoy the same benefits. Super PACs did not have any statistically significant effects on female Democrats' probabilities of existing, entering, and winning. My analysis does not explain the mechanism through which female Republicans benefit from super PACs.

Non-White Democratic candidates are the worst victims of super PACs. Super PACs had negative, statistically significant effects on three of the four variables for non-White Democrats. Super PACs lowered their probabilities of existing, entering, and winning unconditional on running. Unlike non-White Democrats, non-White Republicans did not suffer the same disadvantages. Super PACs did not have any statistically significant effects on non-White Republicans' probabilities of existing, entering, and winning.

Above results support my three hypotheses. My first hypothesis is that super PACs reduce minority representation. My analysis either on both partisan and non-partisan contests or on partisan contests only shows that super PACs decrease the probabilities of minority existing and entering. Super PACs increase the probability of minority winning conditioning on minority running, which implies that super PACs impose a higher barrier on minority candidates, allowing only extremely talented minority candidates to exist or enter. Therefore, I conclude that my first hypothesis holds.

My second hypothesis is that the negative effect is larger for non-partisan local elections. Excluding non-partisan elections from the main analysis made many subgroup effects insignificant. For example, female Democrats received statistically significant effects on their probability on entering and winning conditioning on running based on all local contests as shown in Figure 1. Excluding non-partisan elections made these effects insignificant as shown in Figure 2. Therefore, I conclude that my second hypothesis holds.

My third hypothesis is that the reduction in representation is bigger for non-white candidates than for female candidates. Figure 1 and Figure 2 show that non-White candidates are the worst victims of super PACs among minority whether in partisan or non-partisan elections. The figures also show interesting dynamics between gender and partisanship. Figure 1 shows that super PACs reduce female Democrats' probabilities of entering and increase their probabilities of winning conditioning on running, but the same effects are not found on female Republicans. Additionally, Figure 2 shows that female Republicans' probability of winning conditional on running increases without any change in their probabilities of existing and entering. Thus, I conclude that my third hypotheses holds.

In sum, super PACs have a negative impact on minority representation in local elections. The impact is larger in non-partisan elections with low salience and smaller campaign finance where the influence of a handful of wealthy donors is large. Super PACs have different sizes of effects on different subgroups of minority. Non-White candidates suffer most in both partisan and non-partisan elections. Among female candidates, Democrats suffer more than Republicans. Female Republicans enjoy a slight bump in their probability of winning conditioning on running. Combining subgroups experiencing differential effects into a generic group - minority candidates - cancels out the effects in opposing directions and yield insignificant effects. This result highlights the importance of heterogeneous treatment analysis.

Robustness Checks

A key assumption in DiD models is the parallel trends assumption. This assumption states that, in the absence of the treatment, the treated group and the control group would have followed parallel paths or trends in their outcome variables over time.

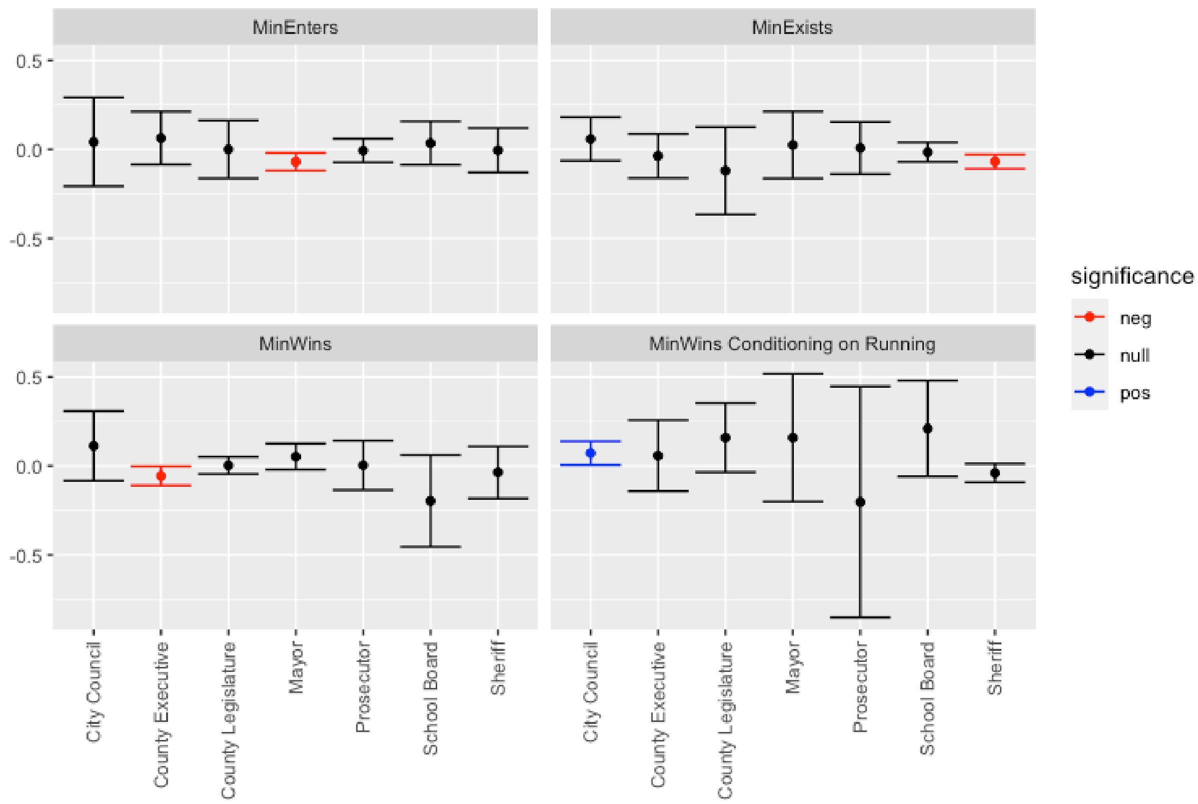

Figure 3: Effect of super PACs by office. The Y-axis indicates the log-odds ratio from the model. The X-axis indicates the type of minority candidates. From left to right, *Fem* indicates female candidates. *Dem* indicates Democratic candidates. *NW* indicates non-White candidates. *Min* indicates minority candidates, which include both female and non-White candidates. *Rep* indiciates Republican candidates. The black dots represent point estimates, and the error bar shows the 95 percent confidence interval. The blue error bars indicate positive and statistically significant effects. The red error bars indicate positive and statistically significant effects.

If the parallel trends assumption does not hold, the DiD estimate may be biased, as the treated and control groups might have followed different trajectories even without the treatment. In such cases, the difference between the pre-treatment and post-treatment outcomes for the two groups could be due to factors other than the treatment itself, making it difficult to isolate the true causal effect of the treatment.

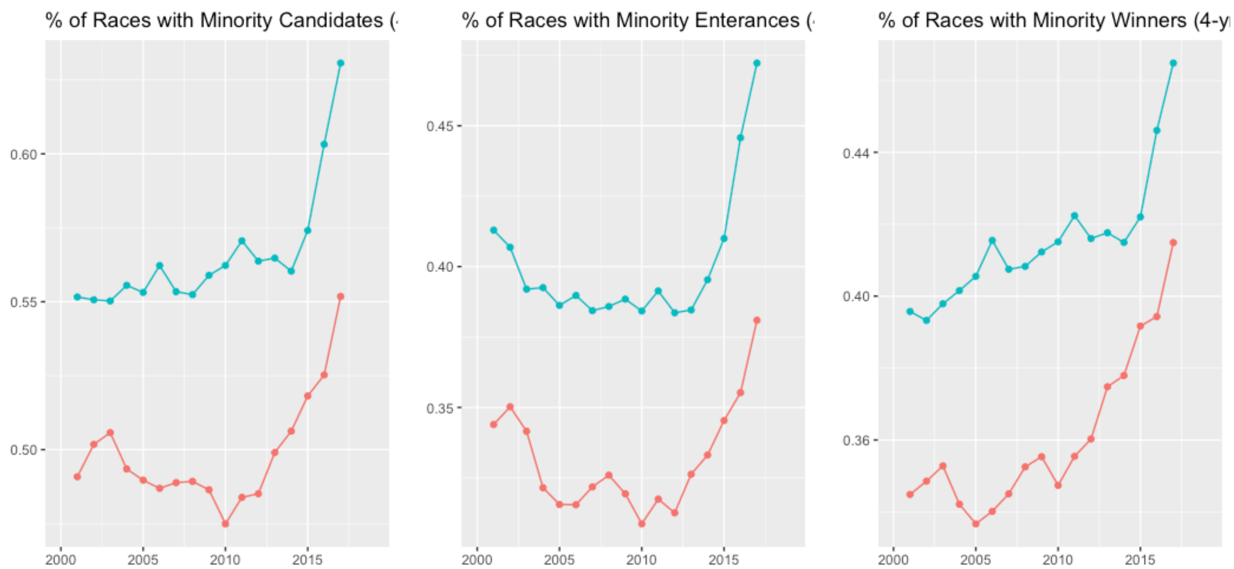

Figure 4: Parallel trend assumption. The X-axis shows years. The Y-axis shows the share of races with minority candidates, minority candidate entrances, and minority winners, respectively. The red line indicates the average value for local elections in treated states, or states that previously had a ban on corporate or union independent expenditures. The blue line indicates the average value for local elections in untreated states, or states that previously did not have a ban on corporate or union independent expenditures.

Figure 4 shows the trend of each of the three outcome variables - *MinExists*, *MinEnters*, and *MinWins* - for treated and untreated groups. The blue line indicates the average value for local elections in treated states, or states that previously had a ban on corporate or union independent expenditures. The red line indicates the average value for local elections in treated states, or states that previously had a ban on corporate or union independent expenditures.

The figure shows that states that did not have a ban on corporate or union independent expenditures have greater minority representation in local elections. For all three outcome variable, the treated group saw a slight dip right before 2010, while the untreated group saw a slight increase.

Such dips may create an upward bias.

Given more time, I would love to run the same analysis with a Generalized Synthetic Control (GSC) design. GSC is a variant of synthetic control method, which are a quasi-experimental method often used when the parallel trends assumption does not hold. GSC method differs from the classic synthetic control method in that GSC handles multiple treated units. GSC uses a weighted combination of untreated units for each treated unit, allowing for flexible counterfactual construction. If my analysis with a GSC design yields similar findings as DiD, that means my findings are robust to method selection.

Limitations

Given time constraints, my preliminary analysis have several limitations. First, I have not yet checked how comprehensive the American local government elections database is. Although this is the most comprehensive publicly available database of local elections across the country, it certainly lacks data for elections held in many smaller localities without the infrastructure to report their election data. In the paper, the authors provides Figure 5 that shows the coverage of their data, but not statistics on the coverage. One key observation is that the data covers only a small subset of school districts in West and East coasts. Given more time, I would love to derive some descriptive statistics on the coverage of this database.

Another limitation mentioned earlier is that my preliminary analysis does not address the issue of redistricting. In the analysis, I completely ignore the issue of redistricting and assume district boundaries remain the same. Given more time, I would love to identify and exclude districts that saw significant boundary changes.

Third, my preliminary results are in log-odds ratios, which are not easily interpretable. Given more time, I would love to transform log-odds ratios to more intuitive probabilities. Additionally, I interpret the finding that super PACs' positive effect on the probability of minority winning conditioning on minority running as evidence that shows that super PACs create so great barrier for

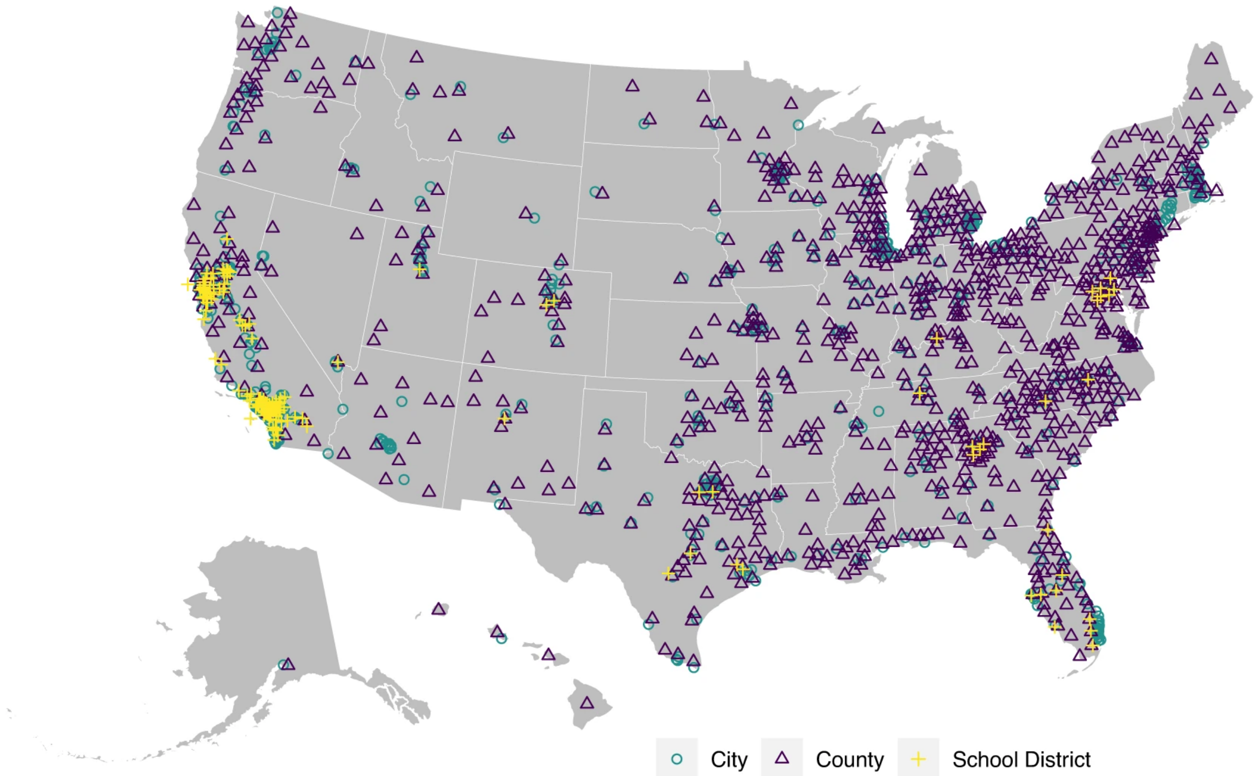

Figure 5: Map of American local government elections data coverage

minority candidates that only extremely qualified ones enter the race. Given more time, I would love to provide supporting evidence by measuring the quality of candidates.

The last limitation is that it is unclear whether the key assumption for my main analysis holds. A graphical analysis in Figure 4 shows that the treated group saw slight dips in outcome variables right before the treatment assignment in 2010. Such dips are likely to create an upward bias in the estimates. Given more time, I would love to run the same analysis with a GSC design, which does not have parallel trend assumption.

Contributions

Despite aforementioned limitations, this paper makes several contributions at the intersection of the literatures on super PACs and local elections. To my knowledge, this is the first paper studying the effect of super PACs on local elections, which became available thanks to the recent release of

American government local elections database by Benedictis-Kessner et al. (2023). My preliminary analysis yields several interesting insights. First, super PACs reduce minority representation. Second, the size of reduction is larger in non-partisan elections with smaller campaign finance where the influence of wealthy donors is larger. Third, super PACs have heterogeneous effects on different subgroups of minority. Non-white minorities are the worst victims while female Republicans slightly benefited.

Conclusion

This study reveals that super PACs significantly undermine minority representation in U.S. local elections. The findings confirm that Super PACs decrease the likelihood of minority candidates existing, entering, and winning elections, with the most substantial barriers observed in non-partisan contests. The results also highlight subgroup-specific disparities, demonstrating that non-White candidates are disproportionately disadvantaged compared to female candidates, with non-White Democrats experiencing the greatest setbacks. Conversely, female Republicans uniquely benefit from the rise of Super PACs, enjoying increased probabilities of winning conditional on running.

These findings underscore the complex interplay between campaign finance, race, and gender in local electoral dynamics. The heterogeneity of effects emphasizes the necessity of nuanced analyses that go beyond aggregated "minority" categories to account for intersectional differences. Moreover, the study's methodological rigor, including robustness checks and the proposed use of Generalized Synthetic Control (GSC) methods, strengthens the validity of its conclusions.

Future research should address the limitations posed by redistricting and explore the mechanisms underlying the observed disparities, such as candidate quality and donor preferences. Policymakers should consider these findings when evaluating the broader societal implications of campaign finance laws, particularly their impact on equity in political representation.

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